

Operations Research

Lecturer: Dr. Pascual Reusch

ALTERNATIVE

“Final Project Operations Research”

Student:

Bárbara Calderón Ávila - FINISHED info * (GREEN)

Jackson

Emmanuel

Marko

Abde

IF YOU AGGREGATE MORE INFO PLEASE HIGHLIGHT IT IN THIS COLOR AND
ADD A COMMENT WITH YOUR NAME so we can discuss it later.

***Python + Heuristic approach needed.**

HELP IN THIS *Jackson and Emmanuel

***DO YOUR PART BY JULY 1ST-SO WE CAN
CHECK TOGETHER THE OTHER ELEMENTS***

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Structure

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CHECKLIST PROF (VERIFY WE HAVE EVERYTHING BY JULY 1ST)

- ☐ A clear OR decision question
- ☐ A real-world motivation
- ☐ A data model with realistic small, medium, and large instances
- ☐ Real data sources or a detailed artificial data scenario
- ☐ Plausibility checks for generated or collected data
- ☐ A mathematical model with variables, objective, and constraints
- ☐ A professional solver implementation
- ☐ Solver status, runtime, objective value, and gap/bound where applicable
- ☐ A meaningful heuristic implementation
- ☐ A direct comparison between solver and heuristic
- ☐ Visualizations that explain the problem and the results
- ☐ An interpretation of the results as decision support
- ☐ A clear recommendation linked back to the real-world motivation
- ☐ A discussion of scalability and why the solver struggles
- ☐ A discussion of practical limitations
- ☐ A Jupyter Notebook and a PDF export with visible outputs

I. Introduction

Lidl is one of the largest and most influential discount supermarket chains in Germany, operating an extensive retail network characterized by high-frequency replenishment cycles. Market reports indicate that Lidl consistently maintains a leading market share in the German discount retail sector, generating tens of billions of euros in annual revenue (Discount Retail Consulting GmbH, 2025). The vast geographic distribution of these retail outlets with high inventory turnover, introduces substantial logistical complexities for suppliers and distribution networks.

Concurrently, Dr. Oetker operates as one of Germany's premier food manufacturers, distributing a diverse product portfolio to supermarkets and retail hubs nationwide within this supply chain. Therefore, highly efficient transportation planning is crucial to guarantee delivery reliability, maintain on-shelf availability, and mitigate operational issues that can lead to increased costs, time delivery and unnecessary *CO2* emissions.

In regional distribution networks, particularly within the Ostwestfalen-Lippe (*OWL*) area, logistical planning is further constrained by strict operational requirements and *stochastic factors** (App.1), including urban traffic congestion and driver-related ergonomic considerations that must be accounted for to ensure efficient service delivery. In this context, heterogeneous vehicle fleets introduce additional complexities such as varying payload capacities, working-time regulations, and customer-specific delivery time windows. These characteristics reflect realistic operational conditions commonly encountered in practice, and they motivate the formulation of more complex mathematical models within the field of Operations Research (OR), as well as the development of efficient algorithmic implementations for their solution, consequently, resulting in a more precise interpretation and analysis to find the optimal solution to the problem.

The Ostwestfalen-Lippe (*OWL*) area* (App.2) is considered as a representative case study for the analysis of complex logistics operations. In this context, logistical planning is subject to a variety of operational constraints and stochastic factors, including urban traffic congestion and driver-related ergonomic considerations, all of which must be accounted for to ensure efficient and reliable service delivery. Moreover, heterogeneous vehicle fleets introduce additional complexity through varying payload capacities, working-time regulations, and customer-specific delivery time windows; otherwise, if these factors are not considered, it may lead to inefficient routing directly: prolonged delivery times, inflated operational overhead, and elevated environmental emissions.

These characteristics reflect realistic operational conditions and provide a suitable framework for formulating a richer and more constrained optimization model within the field of *OR*. The incorporation of such restrictions leads to a more comprehensive mathematical formulation and supports the development of more advanced solution approaches, allowing the problem to be addressed in a way that is closer to real-world decision-making scenarios. At the same time, this formulation serves as an application of the concepts and methodologies

acquired during the course, enabling their integration in a complex, realistic optimization setting.

Furthermore, logistics planning for this case study cannot be isolated of selecting the shortest path between a centralized depot and an individual store. Instead, it requires the design of coordinated multi-stop routes for a heterogeneous fleet while simultaneously considering multiple and often conflicting operational objectives, such as cost efficiency, service reliability, and resource utilization. As the number of customers and feasible routing combinations increases, the problem size grows combinatorially, significantly increasing computational complexity and making manual planning approaches unsuitable for by-hand problem solving; hence, this present document will also show the use and analysis by computational solvers.

To address these challenges, this project develops a sustainability-oriented logistics optimization analysis. By evaluating how mathematical routing decisions can reduce unnecessary travel distance, fuel consumption, and greenhouse gas emissions, the resulting optimization model serves as a robust decision-support system. It enables logistics planners to identify strategic delivery sequences and fleet configurations that maximize operational efficiency without compromising service quality.

Project Question

Which delivery sequence and vehicle capacity configuration (low-, medium-, or high-capacity vehicles) should be selected to distribute Dr. Oetker products to the 24 Lidl stores in the Ostwestfalen-Lippe region in order to minimize transportation time and operational costs while satisfying delivery time windows, vehicle capacities, and ergonomic constraints?

Main objective

The objective of the model is to minimize total transportation and delivery time, while improving distribution efficiency and reducing operational costs associated with vehicle usage and driver allocation.

II. Literature Review

2.1. Classification within the Operations Research Domain

This project addresses a combinatorial optimization problem within the field of *Rich Vehicle Routing Problems (RVRPs)*, a class of models introduced in the *Operations Research (OR)* literature to represent routing problems that extend beyond the assumptions of the classical *Vehicle Routing Problem (VRP)*; unlike traditional *VRP* (first introduced by Dantzig and Ramser in 1959 in the “Traveling Salesman Problem”), *RVRPs* incorporate multiple real-world constraints, such as strict delivery time windows, limited vehicle capacities, ergonomic considerations (for example driver-service ergonomic related constraints), multiple depots, customer priorities, and traffic or road restrictions (Cáceres Cruz, et al. 2014). By considering

this practical focus, *VRP* models enable the development of mathematical formulations and solution algorithms that more accurately reflect real distribution systems, producing solutions that are both operationally feasible and applicable in industrial settings.

Specifically, this project focuses on the *Heterogeneous Fleet Vehicle Routing Problem with Time Windows (HFVRPTW)*, a variant of the *VRP* that considers fleets composed of vehicles with different capacities and operational characteristics while satisfying customer time-window constraints. The proposed mathematical model was developed following an extensive review of the relevant *OR* literature, with particular emphasis on studies addressing heterogeneous fleets and time-constrained routing problems described in the bibliography section.

In high-frequency resupply networks, such as the distribution system analyzed in this present document, ignoring differences in vehicle capacities and urban accessibility constraints may lead to substantial operational inefficiencies like poor vehicle utilization, increased transportation costs, unnecessary trips, violations of delivery time windows, and the assignment of vehicles that are unable to access certain delivery locations due to road regulations, low-emission zones, weight restrictions, or limited urban infrastructure. As a result, the simplifying assumptions of the classical *VRP* is scarce to represent the operational complexity of modern distribution systems, making richer formulations such as the *HFVRPTW* a more appropriate modeling framework.

Furthermore, with a mathematical approach, the *HFVRPTW* is part of a *NP-hard formulations* according to (Gendreau et al., 1999) since it amalgams two type of sub-combinational problems of exponential scale:

1. Bin Packing Problem (BPP): Decide with sub-set of clients and demands are assigned to which sub-set of clients for each demand without exceeding its volumetric or maximum weight capacity (Q_k).
2. Traveling Salesman Problem with Time Windows (TSPTW): Determine the optimal sequence of visit (arc-sequence) for each activated vehicle guaranteeing that the time arrival (t_{ik}) is in the correct time interval for each of the nodes of the system.

2.2 Main pillars of this *HFVRPTW* case study model

- Time Windows

The formal integration of temporal variables and restrictive service time windows in vehicle routing was consolidated by Solomon (1987), whose work *Algorithms for the Vehicle Routing and Scheduling Problems with Time Window Constraints* remains a key reference in *OR* literature for benchmarking routing algorithms. In his study, Solomon showed that introducing time windows significantly reduces the feasible solution space. However, even though this reduction might suggest a simpler problem, it actually introduces additional computational difficulty when solving it with exact or heuristic methods. In particular, determining whether a feasible solution exists becomes more complex due to the mismatch between spatial proximity and temporal compatibility: two customers that are geographically

close may still be incompatible within the same route if their service time windows do not overlap.

- Heterogeneous Fleet and Cost Offsets

Some authors like Gendreau et al. (1999) and, subsequently, Baldacci et al. (2008) formalized the Heterogeneous Fixed Fleet VRP (*HFFVRP*) and the Fleet Size and Mix VRP (*FSMVRP*), respectively. In their work, they showed that the objective function of a realistic *OR* model cannot be restricted to exclusively minimizing total travel distance alone. Instead, it must capture a fundamental cost trade-off structure.

The authors demonstrated the existence of a mathematical trade-off between fixed and variable costs associated with different vehicle types. For instance, deploying a heavy articulated truck involves a high fixed operational cost ($\text{FixedCost}_{\square}$), but benefits from a lower variable cost per pallet-kilometer ($\text{VariableCost}_{\square}$). In contrast, a light rigid vehicle has a significantly lower activation cost, but reaches its capacity limits much earlier, which increases the number of required trips and, consequently, the total variable cost.

- Heuristic algorithms

As computational power increased and gained relevance in *OR* formulations, researchers began to integrate more complex and realistic constraints into routing models. A clear example is Cáceres-Cruz et al. (2014), who mapped the state of the system in *RVRPs*, concluding that the combination of *HFVRPTW* represents one of the most practically relevant frontiers in transportation decision support systems (DSS).

Although their investigations also showed that when these constraints are incorporated into exact mathematical solvers, computational times grow exponentially. This scalability limitation is particularly critical in real-world applications, where decision-making must be performed within short time frames. For this reason, heuristic and metaheuristic algorithms are essential, as they are designed to efficiently explore the solution space and generate high-quality near-optimal solutions in reasonable computational times, especially for instances exceeding 25 to 30 nodes, where exact methods often become computationally intractable.

- Food Logistics and Retail Context in Germany

In order to maintain consistency with the real operational conditions of Dr. Oetker and Lidl considered in this study, Holzapfel et al. (2016) analyzed the distribution structures of discount supermarket chains in Germany. Their study showed that the time windows imposed by retailers such as Lidl are typically “hard time windows” (strict delivery time intervals that must be satisfied, otherwise the solution is infeasible) due to the high level of synchronization required at unloading docks (cross-docking operations), as well as German regulations on driving and rest times for professional drivers (*Fahrpersonalverordnung – FPersV*).

Furthermore, they confirmed that fleet heterogeneity is not merely a modeling assumption but a physical requirement in practice, since certain urban centers restrict the access of trucks exceeding 12 tons during daytime hours. This restriction forces the use of smaller commercial vehicles for last-mile distribution or requires transshipment operations between heavy and light-duty vehicles.

2.3. Modeling Approaches in Rich Vehicle Routing

The formulation of the *Heterogeneous Fleet Vehicle Routing Problem with Time Windows (HFVRPTW)* in the literature generally follows one of three primary mathematical modeling approaches, depending on the specific characteristics of the decision-making process:

- Three-Index Formulation (Arc-Based Models): This framework is widely recognized as the standard extension of the classical vehicle routing problem when dealing with non-homogeneous assets (Irnich et al., 2014). This model relies on a binary decision variable x_{ijk} , which equals 1 if vehicle k directly traverses the directed arc (i,j) connecting two physical nodes within the spatial network, and 0 otherwise.
 - To couple spatial routing with temporal constraints, this approach incorporates continuous tracking variables, traditionally defined as t_{ik} representing the exact arrival time of vehicle k at node i .
 - This approach has the advantage that when introducing vehicle index k it forces the decision space to scale quadratically with the number of customers and linearly with the total fleet size, leading to an overall variable complexity.
 - Matrix Relaxation Weakness: Subtour elimination constraints and time window integrations are explicitly modeled via extensive Big-M formulations. These linear continuous relaxations are notoriously loose. Consequently, standard commercial mixed-integer linear programming (MILP) solvers struggle to prune the branch-and-bound tree effectively, making this approach computationally impossible for instances scaling beyond 20 to 30 customer nodes. However, it remains highly favored in the literature when exact operational control over specialized vehicle properties on explicit arcs is non-negotiable.
- Two-Index Formulation: To mitigate the severe coordinate-space expansion of three-index formulations, modern adaptations frequently collapse individual vehicle tracking by aggregating the heterogeneous fleet into distinct, homogeneous vehicle classes or types (Toth and Vigo, 2014). The principal binary decision variable is reduced to x_{ijm} , which denotes whether any vehicle belonging to class m traverses the arc (i,j) . Instead of relying on continuous time variables per vehicle, subtour elimination and capacity utilization are handled implicitly by transforming the routing architecture into a multi-commodity network flow problem. Here, continuous variables represent the flow of physical load units or accumulated elapsed time along the arcs. While this drastically compresses memory requirements and reduces the overall integer variable count to $O(N^2M)$ (where the number of classes M is

significantly smaller than the total fleet size K), it introduces complex multi-commodity tracking constraints. Furthermore, managing individual vehicle fixed activation costs (FixedCost k) becomes a complex task if a specific class contains a finite, fixed number of assets rather than an infinite pool, often requiring additional cutting planes to prevent the over-allocation of specialized vehicles.

- Set Partitioning Formulation (Route-Based Models): This paradigm fundamentally changes the decision domain by shifting the optimization from arc selections to entire path trajectories.
 - Let Ω represent the complete set of all feasible routes that a heterogeneous vehicle type can execute while completely satisfying capacity limits and hard time windows. The model utilizes a binary decision variable y_r , which equals 1 if route $r \in \Omega$ is selected in the final logistics plan, and 0 otherwise. This structure shifts the mathematical complexity from the solver's branch-and-bound tree to the route-generation phase, serving as the foundation for modern exact decomposition methods.

2.4 Typical Constraints in Real-World Applications

While the classical Vehicle Routing Problem (VRP) simplifies outbound logistics by assuming unrestricted urban access, homogeneous fleets, and flexible delivery timing, industrial distribution networks must integrate a multi-layered matrix of operational and legal restrictions to simulate reality in the best possible way. The literature classifies these into several categories:

- Temporal Constraints: These include Hard Time Windows, where early arrival forces waiting and late arrival causes immediate infeasibility (Bräysy and Gendreau, 2005), and Soft Time Windows, where violations are permitted but penalized financially within the objective function to evaluate cost-service trade-offs (Figliozzi, 2010).
- Physical and Infrastructure Compatibility: Vehicles face compatibility constraints based on customer locations and urban regulations (Cáceres-Cruz et al., 2014). Certain retail centers or urban zones restrict access based on maximum weight (e.g., vehicles over 12 tons), vehicle length, or strict emission standards in Low Emission Zones (Holzapfel et al., 2016). Additionally, specific products might require specialized fleet assets, such as refrigerated compartments for perishable food logistics.
- Driver and Labor Regulations: Real applications must respect legal frameworks regarding working hours. In Germany and the European Union, this involves strict compliance with driving and rest time allocations regulated by the Fahrpersonalverordnung – FPersV (Holzapfel et al., 2016), which directly impacts route duration constraints and maximum allowable driving shifts (Goel, 2009).

2.5 Exact Solver Approaches in the Literature

In modern Operations Research literature, the most prominent exact methods include:

- Branch-and-Cut (B&C): An exact methodology where valid inequalities (such as subtour elimination, capacity cuts, or knapsack-like constraints) are dynamically added to tighten the linear programming relaxation, significantly reducing the search space before branching (Pecin et al., 2017).
- Branch-and-Price (B&P): A framework that applies Column Generation to the Set Partitioning formulation (Feillet, 2010). Since the number of possible feasible routes is exponential, the algorithm starts with a small subset of routes and solves a pricing sub-problem—usually modeled as a Shortest Path Problem with Resource Constraints (SPPRC)—to dynamically inject profitable routes into the master problem.
- Branch-and-Cut-and-Price (BCP): Currently representing the state-of-the-art boundary for exact solvability in heterogeneous routing, this hybridized approach combines the dynamic generation of both columns (routes) and rows (cutting planes), allowing exact solvers to tackle complex instances up to roughly 100 nodes (Baldacci et al., 2008; Choi et al., 2020).

2.6 Heuristic and Approximate Approaches

Because the HFVRPTW belongs to the class of NP-hard formulations, exact solvers often fail to find solutions within reasonable operational timeframes when instances scale past 30–50 nodes (Cáceres-Cruz et al., 2014). To circumvent this, researchers utilize heuristic and metaheuristic frameworks designed to find high-quality, near-optimal solutions efficiently:

Classical Heuristics (Constructive & Improvement) Modified Savings Frameworks: Modern adaptations of constructive algorithms alter the traditional "routing savings" calculations to dynamically incorporate heterogeneous capacity thresholds and fixed vehicle activation costs simultaneously (Coelho et al., 2016).

Time-Oriented Insertion Heuristics: These algorithms sequentially insert unassigned customers into existing routes based on a weighted combination of spatial minimization and temporal delay minimization tailored for tight windows (Bräysy and Gendreau, 2005).

Metaheuristics (Local Search Frameworks) Large Neighborhood Search (LNS) and Adaptive LNS (ALNS): Widely considered the gold standard for rich VRP variants (Pisinger and Ropke, 2007). ALNS works by iteratively destroying a portion of the current solution (removing customers using heuristics like random, worst, or radial removal) and rebuilding it using repair heuristics (Ropke and Pisinger, 2006). The algorithm adaptively learns which destroy/repair pairs perform best during the search.

Variable Neighborhood Search (VNS) and Tabu Search: These methods navigate local search spaces by systematically changing neighborhood structures (e.g., swapping nodes between

routes) while maintaining memory logs or structures to prevent the algorithm from getting trapped in local optima (Bräysy and Gendreau, 2005; Archetti et al., 2014).

Hybrid Genetic Algorithms: Population-based metaheuristics that evolve a set of routing solutions over generations using advanced crossover and mutation operators, frequently hybridized with local search techniques to fix time window and capacity violations dynamically (Vidal et al., 2013).

2.7 Data Requirements for Model Implementation

To successfully operationalize an HFVRPTW model, a comprehensive dataset of spatial, temporal, fleet, and demand characteristics is required.

Specific Parameters Required for each data Category

- *Network & Spatial Data:** Latitude/Longitude coordinates of all nodes; asymmetric travel distance matrices; asymmetric travel time matrices (Cáceres-Cruz et al., 2014)
- *Demand Data:** Deterministic demand volume per customer (d_i) measured in pallets, weight, or volume units matching fleet capacities ((Koç et al., 2016).
- *Temporal Data:** Hard or soft time window intervals ($[e_i, l_i]$) for each customer; service/unloading time required at each node (s_i) (Bräysy and Gendreau, 2005).
- *Fleet Configurations:** Number of available vehicles per category k ; volumetric/weight capacity (Q_k); fixed activation cost ($FixedCost_k$); variable cost per kilometer ($VariableCost_k$) (Baldacci et al., 2008).

2.8 Problem Simplifications and Operational Acceptability

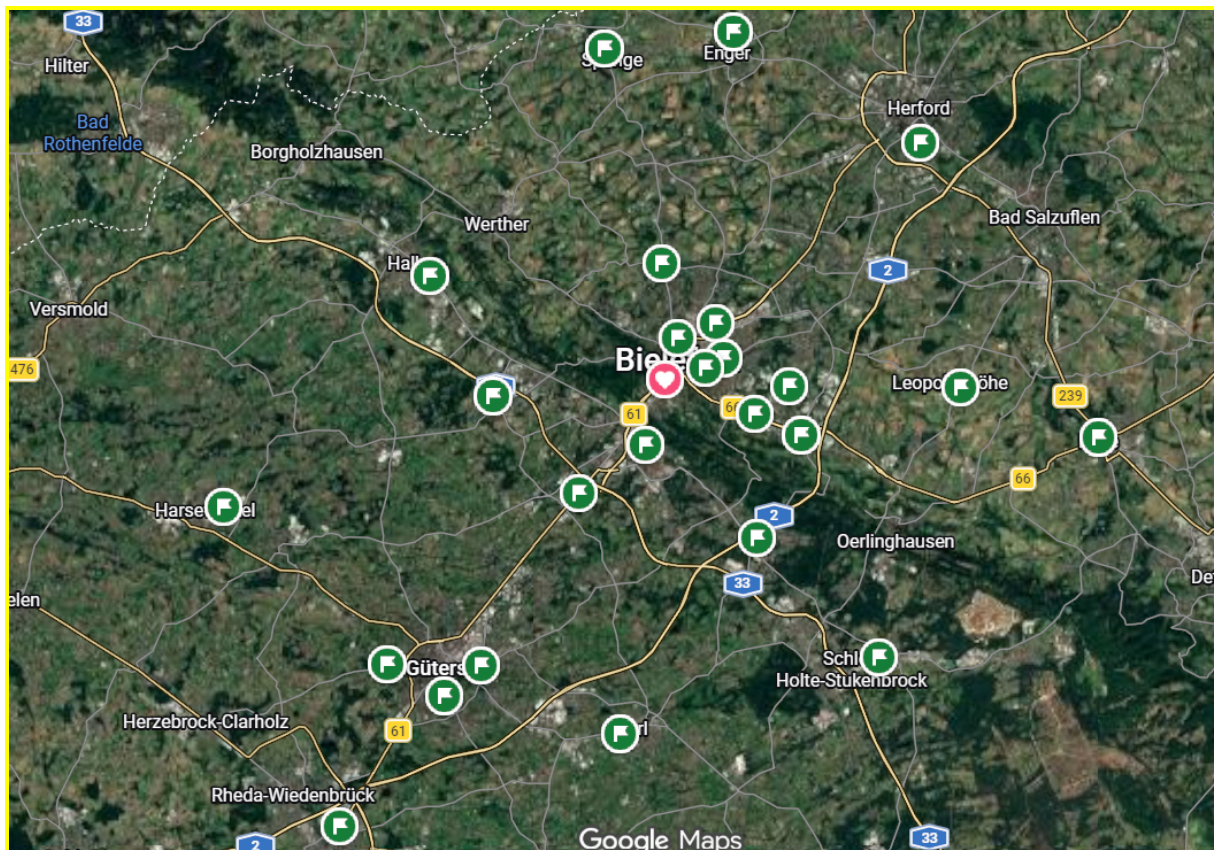
To ensure the mathematical model developed in this project remains computationally feasible while still providing highly valuable decisions for the distribution networks of Dr. Oetker and Lidl, certain simplifications to the real-world problem were applied. Following established Operations Research literature, these assumptions are explicitly stated and justified from an operational and structural perspective:

***Deterministic Travel Times and Demand:** The model treats travel times between nodes and customer demands as constant, known parameters, omitting real-time traffic stochasticity or sudden demand fluctuations. In high-frequency German retail supply chains, demand is heavily stabilized via advanced Electronic Data Interchange (EDI) systems, meaning order quantities are fixed before routing execution begins (Holzapfel et al., 2016). Furthermore, travel times are calculated using historical averages that incorporate standard urban delays, making a deterministic approximation highly reliable for day-ahead tactical planning (Cáceres-Cruz et al., 2014). In order to simulate traffic congestion, it was applied a security factor according to the OWL region (shown in data collection), this ensures that we can run the model and cover the demand in different traffic scenarios, and also, it gives us the possibility to consider different type of vehicles sizes used depending on the time of the day.

Homogeneous Service Times: Unloading and service times (s_i) at customer nodes are calculated based on average standard durations rather than fluctuating dynamically based on specific crew sizes or unexpected dock delays. Large retail actors like Lidl operate under highly synchronized cross-docking and logistics standards (Holzapfel et al., 2016). Dock scheduling slots are strictly regimented, forcing service times to conform tightly to planned averages, which justifies treating them as deterministic parameters in the temporal constraints.

Single-Period Planning Horizon: The routing optimization is executed for a single operational shift or day, without modeling multi-period vehicle re-allocation over a weekly horizon. High-frequency grocery and food resupply networks, such as those governing perishable or ambient food distribution, operate on a daily turnaround cycle due to product freshness and strict continuous replenishment strategies (Holzapfel et al., 2016). Optimizing on a daily rolling horizon matches the true organizational structure of the dispatching teams without introducing the exponential variable growth of multi-period formulations.

III. Data Collection



i\j	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	
0	0	0	13	14	9	16	16	15	16	13	9	21	9	28	26	30	25	27	32	25	20	17	29	31	33	30	32
1	0	13	0	16	7	12	20	19	15	8	18	21	10	31	29	34	29	31	34	28	23	20	24	32	31	28	30
2	14	16	0	13	20	20	22	23	17	11	18	17	26	24	28	23	25	28	23	18	14	25	23	31	27	27	
3	9	7	13	0	9	18	17	17	10	13	16	11	31	29	33	28	30	33	28	23	20	21	29	29	24	27	
4	16	12	20	9	0	27	26	22	18	22	26	19	31	28	32	28	30	33	27	23	19	21	25	29	22	22	
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25	32	30	27	27	22	32	22	36	31	29	37	32	47	45	51	46	45	43	41	29	33	31	12	33	17	0	

time matrix

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	AA
i\j	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
0	0	5.1	6.5	2.9	6.3	7.1	7.1	7	3.7	3.7	10.7	25	28.5	17.1	20.2	21.1	34	23	23.7	14.5	11.2	16.5	16.5	21.6	23.4	20.8
1	5.1	0	11.3	2.9	4.9	7.9	7.8	6	3.8	8.8	19.9	3.1	29.3	18.3	21.3	21.9	35.2	24.2	24.4	15.3	12.4	13	15.9	20.7	17.2	19.4
2	6.5	11.3	0	9.2	12.6	20.8	11.8	13.5	12.2	4.5	10.9	8.9	29.1	17.8	20.8	21.7	34.6	33.6	24.3	15.1	11.8	14.7	13.7	22.7	17.4	18.8
3	2.9	2.9	9.2	0	3.5	8.9	8.9	3.5	6.4	17.4	2.3	31.7	20.3	23.3	24.3	37.2	36.2	26.9	17.7	14.4	12.1	16.6	20.4	16	19.2	
4	6.3	4.9	12.6	3.5	0	12.3	12.3	11	6.8	9.8	20.8	6.4	31.6	20.2	23.2	24.2	37.1	36.1	26.8	17.6	14.3	12.7	14.8	20.6	14.8	17.3
5	7.1	7.9	20.8	8.9	12.3	0	2.4	3.7	6.6	9.8	7.2	6.1	27.9	26.7	31.5	27.6	33.4	36.2	16.5	17.8	14.5	13.3	19.7	17.4	19.2	32.8
6	7.1	7.8	11.8	8.9	12.3	2.4	0	2.6	4.2	6.7	10.2	3.7	32.5	21.1	24.2	25.1	38	30	27.7	15.9	13.7	15.6	10.7	25.8	16.9	13.7
7	7	6	13.5	8.9	11	3.7	2.6	0	3.8	8.8	11.2	4.2	16.2	13.8	16.9	15.1	31.2	21.4	15	13.6	10.3	19	21.5	23.3	28.7	25.7
8	3.7	2.1	12.2	3.5	6.8	6.6	4.2	3.8	0	8.5	11.7	1.4	27.2	25.9	30.7	26.9	32.7	37.6	15.8	31.2	24.6	8.8	20.6	16.7	18.5	32
9	3.7	8.9	4.5	6.4	9.8	9.8	6.7	8.8	8.5	0	7.5	5.3	26.1	24.9	29.7	25.8	31.7	36.5	12.7	30.2	23.6	9.6	23.7	15.7	17.5	31
10	10.7	19.9	10.9	17.4	20.8	7.2	10.2	11.2	11.7	7.5	0	11.1	12.6	9.2	12.2	13.1	30.1	17.4	19.8	16.5	7.7	27.6	22.8	31.7	33.3	27.1
11	2.5	3.1	8.9	2.3	6.4	6.1	3.7	4.2	1.4	5.3	11.1	0	19	17.8	22.6	12.6	24.6	35.6	8.5	23.1	16.5	17.1	30.8	21.2	22.8	36.3
12	28.5	29.3	29.1	31.7	31.6	27.9	32.5	16.2	27.2	26.1	12.6	19	0	3.1	4.6	9.5	8.7	17.1	26.2	22	15.6	33.6	37.9	37.7	39.3	51
13	17.1	18.3	17.8	20.3	20.2	26.7	21.1	13.8	25.9	24.9	9.2	17.8	3.1	0	5.7	8.2	10.4	15.6	24.9	20.2	13.4	32.3	31.7	36.5	38	49.7
14	20.2	21.3	20.8	23.3	23.2	31.5	24.2	16.9	30.7	29.7	12.2	22.6	4.6	5.7	0	13	8.7	12.8	29.8	21.4	16.2	37.2	37.1	41.3	42.9	41.2
15	21.1	21.9	21.7	24.3	24.2	27.6	25.1	15.1	26.9	25.8	13.1	12.6	9.5	8.2	13	0	14.3	24.2	13.2	27.7	21.1	29.1	35.5	33.3	34.9	46.5
16	34	35.2	34.6	37.2	37.1	33.4	38	31.2	32.7	31.7	30.1	24.6	8.7	10.4	8.7	14.3	0	20.5	31.1	41.1	34.5	38.5	48.9	42.6	44.2	55.9
17	23	34.2	33.6	36.2	36.1	36.2	30	21.4	37.6	36.5	17.1	15.6	12.8	24.2	20.5	0	41.9	17.1	13.8	49.7	31.7	53.8	49.7	53.8	55.4	35.6
18	23.7	24.4	24.3	26.9	26.8	16.5	27.7	15	15.8	12.7	19.8	8.5	26.2	24.9	29.8	13.2	31.1	41.9	0	29.3	22.7	13.9	47.7	17.9	27.6	39.2
19	14.5	15.3	15.1	17.7	17.6	17.8	15.9	13.6	31.2	30.2	16.5	23.1	22	20.2	21.4	27.7	41.1	17.1	29.3	0	6.9	38.1	17.3	42.2	43.8	21.4
20	11.2	12.4	11.8	14.4	14.3	14.5	13.7	10.3	24.6	23.6	7.7	16.5	15.6	13.4	16.2	21.1	34.5	13.8	22.7	6.9	0	31.3	21	35.5	37	25.4
21	16.5	13	14.7	12.1	12.7	13.3	15.6	19	8.8	9.6	27.6	17.1	33.6	32.3	37.2	29.1	38.5	49.7	13.9	38.1	31.3	0	25.1	8.2	14.2	27.8
22	16.5	15.9	13.7	16.6	14.8	19.7	10.7	21.5	20.6	23.7	22.8	30.8	37.9	31.7	37.1	35.5	48.9	31.7	47.7	17.3	21	25.1	0	32.5	18.7	7.5
23	21.6	20.7	22.7	20.4	20.6	17.4	25.8	23.3	16.7	15.7	31.7	21.2	37.7	36.5	41.3	33.3	42.6	53.8	17.9	42.2	35.5	8.2	32.5	0	16	29.6
24	23.4	17.2	17.4	16	14.8	19.2	16.9	28.7	18.5	17.5	33.3	22.8	39.3	38	42.9	34.9	44.2	55.4	27.6	43.8	37	14.2	18.7	16	0	15.9
25	20.8	19.4	18.8	19.2	17.3	32.8	13.7	25.7	32	31	27.1	36.3	51	49.7	41.2	46.5	55.9	35.6	39.2	21.4	25.4	27.8	7.5	29.6	15.9	0

distance matrix

- Demands for each store

Node	Location	Road Zone Type (Industrial, Open, Historic, Residential)	Type of fleet permitted	Technical Justification
N_0	Lutterstraße 14 · 33617 Bielefeld	Industrial zone	All categories	Designed for standard heavy-duty logistics.
N_1	Teutoburger Straße 98, 33607 Bielefeld-Mitte	Historic / Central urban	Low capacity only	Located in a dense urban center with strict axle weight and geometric turning radius constraints. High/Medium trucks are legally/physically restricted.
N_2	Jöllennecker Straße 41, 33613 Bielefeld-Mitte	Historic / Central urban	Low capacity only	High-density urban main street with restricted loading bays and narrow approach corridors. High-capacity vehicles prohibited.
N_3	Heeper Straße 113, 33607 Bielefeld-Mitte	Residential zone	Low, Medium	Residential/commercial transition zone. Structural noise regulations and traffic calming measures prevent the use of

				high-capacity heavy-duty trucks
N_4	Stadttheider Straße 11, 33609 Bielefeld-Mitte	Industrial zone	All categories	Industrialized commercial area with dedicated wide access roads and high-tonnage docking bays.
N_5	Detmolder Straße 345, 33605 Bielefeld-Stieghorst	Open / Arterial road	All categories	Situated on a major arterial transit road with seamless highway connection. Fully accessible for all vehicle dimensions and capacities.
N_6	Babenhauser Straße 18, 33613 Bielefeld-Schildesche	Residential zone	Low, Medium	Dense residential periphery. High-capacity multi-axle trucks are banned due to low clearance, tight inner-district curves, and local noise protection ordinances.
N_7	Kimbernstraße 1, 33647 Bielefeld-Brackwede	Open road	All categories	Outskirts commercial zone. Broad approach roads allow unrestricted maneuvering for high-capacity trucks without urban bottlenecks.
N_8	Oldentruper Straße 255, 33719 Bielefeld-Heepen	Industrial zone	All categories	Standard industrial park setup. Infrastructure built for heavy freight, including large turning loops and automated docking ramps.
N_9	Detmolder Straße 550, 33699 Bielefeld-Stieghorst	Open / Arterial road	All categories	Direct proximity to the federal highway interface. Exceptional structural accessibility with zero geometric or legislative barriers for the entire fleet.
N_{10}	Kasseler Straße 1, 33649 Bielefeld-Brackwede	Open road	All categories	Regional connecting route with wide lane layouts. Supports maximum gross vehicle weight, enabling full high-capacity truck operations.
N_{11}	Henleinstraße 1, 33689 Bielefeld	Industrial zone	All categories	Heavy-duty commercial grid. Optimized for heavy distribution logistics; no structural scaling down required for high-capacity vehicles.
N_{12}	Neuenkirchener Str. 100, 33332 Gütersloh	Open road	All categories	Major link road in Gütersloh. Generous spatial clearance and high load-bearing capacity for full-scale logistics fleets.
N_{13}	Carl-Bertelsmann-Straße 152, 33332 Gütersloh	Industrial zone	All categories	Proximity to massive corporate industrial complexes. Roadways engineered specifically for high-frequency heavy-tonnage vehicle flow.
N_{14}	Auf dem Stempel 10, 33334 Gütersloh	Residential zone	Low, Medium	Suburban residential hub. Subject to municipal weight caps and street-side parking narrowing, which makes high-capacity truck transit physically hazardous.

N_{15}	Westring 11, 33415 Verl	Open road	All categories	Outer bypass ring road. Free of historical urban cores or narrow street architectures, enabling uninhibited logistics for all vehicle sizes.
N_{16}	Bielefelder Str. 55, 33378 Rheda-Wiedenbrück	Historic/Central urban	Low capacity only	Enclosed historic urban layout with narrow medieval-derived road widths and strict local environmental/vibration laws. Only a low-capacity fleet is allowed
N_{17}	Berliner Ring 40, 33428 Harsewinkel	Open road	All categories	Major ring distribution route. Designed for cross-regional freight transit with expansive spatial lanes and high maneuverability indexes.
N_{18}	Lüchtenstraße 61, 33758 Schloß Holte-Stukenbrock	Residential zone	Low, Medium	Strict suburban residential zoning. Heavy environmental noise-mitigation caps enforce a strict ban on high-capacity, multi-axle freight vehicles
N_{19}	Elsa-Brändström-Str aße 1, 33790 Halle (Westfalen)	Residential zone	Low, Medium	Strict suburban residential zoning. Heavy environmental noise-mitigation caps enforce a strict ban on high-capacity, multi-axle freight vehicles
N_{20}	Bahnhofstraße 23, 33803 Steinhagen	Historic / Central urban	Low capacity only	Core municipal retail street with tight parking clusters and restricted delivery windows. Physical geometries completely block high/medium-capacity units.
N_{21}	Krentruper Str. 34, 33818 Leopoldshöhe	Open road	All categories	Rural/open arterial infrastructure with expansive lanes. No geometric bottlenecks, allowing high-capacity trucks to minimize delivery runs.
N_{22}	Lange Str. 35-37, 32139 Spenge	Historic / Central urban	Low capacity only	Spenge's central corridor has narrow roadside lanes and structural overhangs. High-capacity vehicles are structurally excluded to prevent gridlock.
N_{23}	Friedrich-Petri-Straße 18-26, 32791 Lage	Residential zone	Low, Medium	Medium-density residential neighborhood. Low vertical clearance from trees/cables and tight street parking removes high-capacity vehicle options.
N_{24}	Ahmser Str. 50-54, 32052 Herford	Industrial zone	All categories	High-volume industrial bypass. Built to handle heavy freight traffic; fully compliant with high-capacity truck requirements
N_{25}	Lambernweg 46, 32130 Enger	Open Road	All categories	Unrestricted peripheral connecting road. Large layout allows optimal route consolidation using high-capacity vehicles

				without accessibility penalties.
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Store Classification and Parameterization Logic

The physical category size of a store (determined by its geographic location and urban zoning) directly impacts its backroom storage capacity, the turnover of ambient and frozen products, and (crucially) the type of truck that can access the facility. If a store is located in a narrow historical district where only a small truck (Low) can fit, its physical demand cannot exceed that truck's maximum capacity. Otherwise, the MIP model would become infeasible unless multiple delivery visits were permitted.

Store Categories

- **Group A: Small / Express Stores** (Historical Districts / Dense Urban Centers): Limited space, high turnover, but replenished in small batches.
 - *Estimated Demand: 2 - 5*
- **Group B: Medium / Standard Stores** (Residential Areas / Connector Streets): Average-sized neighborhood Lidl supermarkets.
 - *Estimated Demand: 6 - 12*
- **Group C: Large / Hyper-Lidl or Hub Stores** (Industrial Zones / Open Highways / Arterial Roads): Large suburban commercial complexes equipped with full loading docks and high storage capacity.
 - *Estimated Demand: 14 - 25*

Demand Parameter Profiles & Data Distribution Model (d_j)

The baseline demand is modeled using the following function based on the store's microlocation characteristics:

$$d_j = \text{Footprint Base} + \text{Transit Density Bonus} - \text{Spatial Buffer}$$

To make the formula work, we establish fixed numerical values for the qualitative features of each location category:

1. *Footprint Base*: Tied directly to the store category size.

- Small = 3 pallets (Compact retail square footage, limited backroom cold-storage footprint)
 - Medium = 8 pallets (Standard standalone suburban supermarket layout)
 - Large = 18 pallets (Regional flagship hub with extensive warehouse facilities).
2. *Transit Density Bonus*: Added if the store sits on a high-traffic highway, main commuter street, or commercial corridor (+1 to 6 pallets depending on regional prominence).
- +0: Minor neighborhood road or restricted town pedestrian zone.
- +1: Standard multi-lane city commuter street.
- +2/+3: Major arterial cross-town ring road or federal highway (\$B\text{-Road}\$).
- +4/+6: Direct accessibility to a major federal Autobahn highway interchange (\$A2 / A33\$).
3. *Spatial Buffer*: Subtracted if the store has severe physical restrictions, localized competition, or if the demand is split s in a lower-density municipal zone (-1 to -3 pallets).
- 0: No limiting spatial factors; captured demand is fully sustained.
- 1: Minor spatial restrictions (example: historical core layout slows turnaround)
- 2/-3: Shared municipal market or adjacent direct regional grocery competitors splitting the shopping catchment basin.

This rigorous, systematic translation converts qualitative spatial characteristics of the Ostwestfalen-Lippe retail landscape directly into fixed, reproducible numerical constraints. This enables the model to bypass corporate confidentiality restrictions while providing an auditable data model for the optimization solver.

Lidl Store Size Classification Framework

- Small=S
 - Average Sales Area: $600\text{ m}^2 - 900\text{ m}^2$

- Backroom Warehouse Size (Approximated) = $100m^2$ - $150 m^2$
- Rack Capacity: 2–3 standard industrial rack levels high, strictly limited to a total of 15–25 pallet slots in backroom storage (due to architectural or historical constraints, narrow layouts, or structural building columns).
- Operational Deamdn Range: 2 to 5 pallets per delivery
- Logistical Rationale: These stores rely on a Just-In-Time continuous replenishment loop. Their compact backroom freezers cannot accept bulk freight without causing floor congestion. Physically, they are restricted to low-capacity delivery vehicles (Low fleet only) due to tight inner-city geometric constraints.
- Medium = M
 - Average Sales Area: $1000 m^2$ - $1400 m^2$
 - Backroom Warehouse Size (Approximated) = $200m^2$ - $300 m^2$
 - Rack Capacity: 3–4 heavy-duty industrial pallet rack levels high, storing 60–100 pallet slots overall.
 - Operational Deamdn Range: 2 to 5 pallets per delivery
 - Logistical Rationale: This represents the classic suburban standalone supermarket with its own dedicated surface parking lot and loading area. The warehouse size is optimized to cleanly absorb a standard mixed ambient/cold-chain delivery drop from a medium-sized vehicle without disturbing ongoing stock rotation.
- Large= L
 - Average Sales Area: $1500 m^2$ - $2,200 m^2$
 - Backroom Warehouse Size (Approximated) = $400m^2$ - $600 m^2$
 - Rack Capacity: 4–5 multi-tier industrial storage racks with dedicated drive-in bays, allowing **150–250+ pallet slots** in backroom inventory.
 - Operational Deamdn Range: 14 to 25 pallets per delivery.
 - Logistical Rationale: High-volume peripheral branches strategically located near highways, arterial loops, or light industrial commercial clusters. They are built with fully automated docking ramps and heavy freight loops designed to handle massive, multi-axle heavy-duty trucks loaded to maximum capacity (High fleet permitted).

This physical classification ensures that generated data parameters correspond directly with fleet routing capacities, preventing the mathematical solver from assigning unfeasible bulk shipments to structurally restricted nodes.

Nod e	Location	Road Zone Type	Size Classification	Generated demand (pallets) (di)	Technical Justification
N_1	Teutoburger Straße 98, 33607 Bielefeld-Mitte	Historic / Central urban	S	FootPrint: 3 Transit Density Bonus: 0 (Pedestrian Zone) Spatial Buffer: 0 d1= 3 pallets	Positioned deep within an urban neighborhood center. The compact city square footage prevents high pallet stock intake, and because deliveries must happen through a restricted courtyard, the transit bonus is 0.
N_2	Jöllenbecker Straße 41, 33613 Bielefeld-Mitte	Historic / Central urban	S	FootPrint: 3 Transit Density Bonus: 1 (Commuter Street Bonus) Spatial Buffer: 0 d2 = 4 pallets	Although classified as a small store footprint, it sits on a primary high-density city commuter street out of the central ring, which awards a +1 transit bonus. Rapid shelf-depletion rates drive up the demand by 1 extra pallet compared to N1.
N_3	Heeper Straße 113, 33607 Bielefeld-Mitte	Residenti al zone	M	FootPrint: 8 Transit Density Bonus: 0 (Standard Access) Spatial Buffer:0 d13 = 8 pallets	A classic suburban standalone Lidl supermarket, capturing a standard middle-class housing development demographic. Perfect operational baseline without external density shocks.
N_4	Stadttheider Straße 11, 33609 Bielefeld-Mitte	Industrial zone	L	FootPrint: 18 Transit Density Bonus: 0 (Industrial Baseline) Spatial Buffer:0 d4= 18 pallets	A large-scale branch built inside a commercial/light industrial inner-city pocket.
N_5	Detmolder Straße 345, 33605 Bielefeld-Stieg horst	Open / Arterial road	L	FootPrint: 18 Transit Density Bonus: 4 (Autobahn Entrance Corridor Bonus) Spatial Buffer: 0 d5 = 22 pallets	Sits squarely on the main arterial route leading out toward the B66 and A2 Autobahn entry points. Exceptional transit flow generates massive multi-pallet replenishment requirements (+4 bonus).

N_6	Babenhauser Straße 18, 33613 Bielefeld-Schil desche	Residenti al zone	M	FootPrint: 8 Transit Density Bonus: 0 (Student demographics buffer) Spatial Buffer: -1 7 pallets	Close to university residential sectors. While customer volume is high, the purchase profile leans toward single-person baskets and smaller packaging sizes rather than massive multi-pallet family stocks, triggering a small operational buffer reduction of -1
N_7	Kimbernstraße 1, 33647 Bielefeld-Brac kwede	Open road	L	FootPrint: 18 Transit Density Bonus: 0 Spatial Buffer: -2 d7 = 16 pallets	An expansive retail park location. However, severe morning rush-hour congestion along the industrial access corridors slightly chokes rapid early-day truck turnarounds, applying a -2 spatial buffer adjustment.
N_8	Oldentruper Straße 255, 33719 Bielefeld-Heep en	Industrial zone	L	FootPrint: 18 Transit Density Bonus: 2 (Commercial Hub Axis) Spatial Buffer: 0 d8 = 20 pallets	Embedded into a heavy industrial-commercial axis with flawless multi-lane access roads that support non-stop high-capacity freight distribution (+2 bonus).
N_9	Detmolder Straße 550, 33699 Bielefeld-Stieg horst	Open / Arterial road	L	FootPrint: 18 Transit Density Bonus: 6 Spatial Buffer: 0 d9 = 24 pallets	The absolute flagship store of the network, resting directly at the highway interchange border. It operates as a regional cross-dock shopper hub, earning the maximum +6 transit infrastructure bonus.
N_{10}	Kasseler Straße 1, 33649 Bielefeld-Brac kwede	Open road	M	FootPrint: 18 Transit Density Bonus: 0 Spatial Buffer: 3 (Heavy Local congestion buffer) d10 = 15 pallets	While it is a huge retail facility, it sits on a major southern exit bottleneck that suffers from extreme rush-hour local gridlock, forcing a -3 operational spatial buffer to account for delayed delivery processing.
N_{11}	Henleinstraße 1, 33689 Bielefeld	Industrial zone	L	FootPrint: 18 Transit Density Bonus: 1 (Industrial Park Collector) Spatial Buffer: 0 d11 = 19 pallets	Positioned inside a spacious modern industrial park. Built explicitly for heavy freight turnover with clean geometric docking bays (+1 bonus).
N_{12}	Neuenkirchene r Str. 100, 33332	Open road	M	FootPrint: 8 Transit Density Bonus: 3 (Cross-Town Highway	Situated on a high-speed arterial cross-town highway in Gütersloh. It captures high-volume suburban

	Gütersloh			Bonus) Spatial Buffer: 0 d12 = 11 pallets	car traffic and commuter flows heading back toward outer residential sectors (+3 bonus).
N_{13}	Carl-Bertelsmann-Straße 152, 33332 Gütersloh	Industrial zone	L	FootPrint: 18 Transit Density Bonus: 3 8Corporate Headquarters Corridor) Spatial Buffer: 0 d13 = 21 pallets	Directly adjacent to massive multi-national corporate headquarters and manufacturing campuses in Gütersloh. High worker density creates a huge local consumer pool (+3 bonus).
N_{14}	Auf dem Stempel 10, 33334 Gütersloh	Residential zone	M	FootPrint: 8 Transit Density Bonus: +1 (Residential Collector Road) Spatial Buffer: 0 d14 = 9 pallets	Embedded directly into a massive family housing development. High average cart volumes for bulk-packed household goods justify a minor +1 transit density flow increase.
N_{15}	Westring 11, 33415 Verl	Open road	L	FootPrint: 18 Transit Density Bonus: 0 Spatial Buffer: -1 (Peripheral Delay Buffer) d15 = 17 pallets	Located on Verl's outer high-speed transit ring. It has a spacious footprint, but its remote peripheral distance adds slight transit variances, represented by a small -1 buffer constraint.
N_{16}	Bielefelder Str. 55, 33378 Rheda-Wiedenbrück	Historic/Central urban	S	FootPrint: 3 Transit Density Bonus: 0 (Standard Core) Spatial Buffer: 0 d16 = 3 pallets	Serves a historical smaller city core with lower population density than Bielefeld-Mitte, yielding a steady, basic baseline without local traffic or competition distortions.
N_{17}	Berliner Ring 40, 33428 Harsewinkel	Open road	M	FootPrint: 8 Transit Density Bonus: 4 (Major Regional Bypass Bonus) Spatial Buffer: 0 d17 = 12 pallets	Located on the primary regional bypass ring road. This location captures heavy rural automobile traffic entering and exiting Harsewinkel, justifying a significant +4 transit density increase.
N_{18}	Lüchtenstraße 61, 33758 Schloß Holte-Stukenbrock	Residential zone	M	FootPrint: 8 Transit Density Bonus: 0 Spatial Buffer: -2 (Regional Competitor Split Buffer) d18 = 6 pallets	This node operates in a lower-density municipal zone with close, direct competition from overlapping regional supermarkets on the same street, resulting in a -2 demand split

					buffer.
N_{19}	Elsa-Brändström-Straße 1, 33790 Halle (Westfalen)	Residential zone	M	FootPrint: 8 Transit Density Bonus: 0 Spatial Buffer: 0 d19 = 8 pallets	Positioned optimally next to public sports facilities and residential blocks, preserving a perfectly clean, uninterrupted medium-size store baseline demand.
N_{20}	Bahnhofstraße 23, 33803 Steinhagen	Historic / Central urban	S	FootPrint: 3 Transit Density Bonus: 1 (High Street Flow) Spatial Buffer: 0 d20 = 4 pallets	Located on Steinhagen's main retail high street. Heavy transit passenger flow from neighboring transit lines guarantees a steady inventory drawdown, prompting a +1 transit adjustment.
N_{21}	Krentruper Str. 34, 33818 Leopoldshöhe	Open road	M	FootPrint: 8 Transit Density Bonus: 2 (Commuter Town Feeder) Spatial Buffer: 0 d21 = 10 pallets	Serves an expanding commuter town between Bielefeld and Lemgo. High home-ownership rates mean larger family grocery baskets, warranting a +2 logistical flow bonus.
N_{22}	Lange Str. 35-37, 32139 Spenge	Historic / Central urban	S	FootPrint: 3 Transit Density Bonus: 2 (Regional Feeder Bonus) Spatial Buffer: 0 d22 = 5 pallets	Spenge is a peripheral town core. This store acts as a primary grocery point for multiple surrounding rural settlements, capturing out-of-town car commuters on the main municipal feeder road (+2 bonus).
N_{23}	Friedrich-Petri-Straße 18-26, 32791 Lage	Residential zone	M	FootPrint: 8 Transit Density Bonus: 1 (Municipal Thoroughfare) Spatial Buffer: 0 d23 = 9 pallets	A standard medium municipal branch in the Lippe district capturing regular daily vehicle movements along the city's connecting thoroughfare (+1 bonus).
N_{24}	Ahmser Str. 50-54, 32052 Herford	Industrial zone	L	FootPrint: 18 Transit Density Bonus: 5 (Industrial Corridor Master Axis) Spatial Buffer: 0 d24 = 23 pallets	A massive flagship location anchoring Herford's main industrial logistics corridor. Its extensive cold-chain bays allow it to handle heavy bulk loads seamlessly, yielding a +5 infrastructure bonus.
N_{25}	Lambernweg 46, 32130 Enger	Open Road	M	FootPrint: 8 Transit Density Bonus: 2 (Outer Ring Capture)	Positioned directly on the outer connecting ring road, intercepting peripheral rural

				Spatial Buffer:0 d25 = 10 pallets	traffic prior to entry into Enger's retail center (+2 bonus).
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- Matrix of coordinates for each lidl 1 to 24 and N0, including its direction. TABLE (DISTANCE + TIME)
- Add an image of all stores considered in the region + the Dr. Oetker node (N0).
 - +Include an image of the LIDL stores circled in red that CANNOT be accessed by HIGH capacity trucks.
- Time Windows for each lidl store. JUSTIFY WHY THEY WERE ESTABLISHED LIKE THAT.
 - Error margin considered? Why like that?
 - Residential Quiet-Hour Laws and Local Ordinances (*Lärmschutzverordnung*): In Germany, the unloading of heavy goods in mixed-use or residential areas is strictly regulated or prohibited before 6:00 AM and after 10:00 PM to mitigate noise pollution.
- Average travel speed in the routes considered (does it change depending on the type of vehicle?) , + are there specific routes that have an extra fee to consider?
- Demand pallet assumption per EACH store, and a CLEAR justification of why that was assumed (could be size of the lidl store, location). We already know that if we go to a close LIDL here in Bielefeld is about (15 products) but add a margin depending if it's low/medium/high demand.
 - For example you can research what is the normal consumption of Dr. Oetker per german.
- Demand Assumption buffer depending on season in OWL area. EXPLAIN.
- Traffic security factor considered depending on the time (morning/afternoon/night)
- Investigate which type of fuel is used and price, and how can we decrease C02 emissions in the sequence.
 - C02 emissions generated by type of vehicle.
- Number of available vehicles/drivers by Dr. Oetker (investigate or ASSUME) can offer + JUSTIFY.

- Working Hours and Driving Times in Germany *Number of visits (table below, justify *or even change if you consider it)
- Traffic congestion security factor

The Ostwestfalen-Lippe (OWL) region, centered around Bielefeld, Gütersloh, and Herford, represents one of the most dense economic and industrial clusters in Germany. It serves as a primary logistical corridor connecting western German manufacturing hubs with trans-European freight routes via the A2 and A33 autobahns. Therefore, this network suffers from severe asymmetry: heavy freight transport coexists with intense traffic along critical bottlenecks (such as Ostwestfalenstraße, Detmolder Straße, Herford Straße) and key autobahn interchanges like the **Kreuz Bielefeld** (A2/A33).

In traditional OR, travel times are often assumed to be fixed, although in a real-world scenario like food distribution from the central Dr. Oetker plant, assuming static travel times introduces major scheduling risks. If a vehicle departs during peak commuter hours, ignoring traffic delays leads to time-window violations, resulting in failed deliveries, administrative penalties, and spoiled cold-chain products. Consequently, to better represent commuter and freight congestion patterns on the Bielefeld and wider OWL road network, it was established a “buffer constraint” penalizing or favoring travel times based on the exact departure epoch from the origin vertex:

Key Empirical and Operational Rationales for $\mu(t)$

	Temporal Framework	Regional Infrastructure Stress	Urban Network Validation	Mathematical Purpose
$\mu(t) =$				
$\mu(t) =$				
$\mu(t) =$				

Formally, the congestion multiplier is defined as the piecewise step function:

$$\mu(t) = \begin{cases} 1.3, & 0 \leq t < 120 \\ 1.0, & 120 \leq t < 270 \\ 1.1, & 270 \leq t \leq 300 \end{cases}$$

Vehicle capacity limitations:

Category	Features	Vehicle type	Dimensions (L x W x H)
Low	• Small vehicle. • Low fixed cost. • High maneuverability. • Suitable for stores with low demand.	Motor Vehicle (7.5 t)	6.00 x 2.45 x 2.40 m
Medium	• Typical regional distribution vehicle. • Good balance between capacity and accessibility.	Trailer (18 t)	8.10 x 2.45 x 2.5 m
High	• European standard vehicle with maximum capacity. • Maximum load consolidation. • Lower cost per pallet transported. • Low Maneuverability: This type cannot access to some street due to its size.	Articulated truck (40 t)	13.6 x 2.45 x 2.70 m

Table 1. Vehicle characteristics for each type of category.

Standard Euro Pallet size considered: 80 cm × 120 cm (EN 13698 - 1:2003).

Any solution violating these conditions is considered infeasible.

Capacity measure	Low	Medium	High
Euro pallets (Q_k)	15	20	34
Payload (t)	2.8	13.5	25

Table 2. Vehicle capacity in terms of Europallets and payload for each category type.

Dimensions, payloads, and Europallets retrieved from: (Orbiet Logistik, 2021), considering the Council Directive 96/53/EC.

These specific vehicles were chosen because they exhibit significant fluctuations in an optimization study, allowing the algorithm to select the ideal option across different vehicle sizes. This selection represents a clear leap in capability and reflects distinctly differentiated operational levels.

Example (in terms of euro pallets capacity):

Medium – Low: $20 - 15 = 5 \text{ pallets}$: $5/15$: $+ 33.333\%$

High – Medium: $34 - 20 = 14 \text{ pallets}$: $14/20$: $+ 70\%$

High – Low: $34 - 15 = 19 \text{ pallets}$: $19/15$: $+ 126.667\%$

Far enough apart for the model to reveal differences in:

- Number of routes
- Vehicle utilization
- Adherence to time windows
- Total cost
- Total operating time

The size truck- capacity (Q) is different, $Q_{low} \neq Q_{Medium} \neq Q_{High}$ and affects the next factors directly:

- Gas cost consumption in euros (C): $C_{low} \neq C_{Medium} \neq C_{High}$
- Service time restrictions (D): $D_{low} \neq D_{Medium} \neq D_{High}$
- Maximum store visits (V): $V_{low} \neq V_{Medium} \neq V_{High}$

Factors	Low	Medium	High
C (euros)	~ 0.25 € / km	~ 0.38 € / km	~ 0.56 € / km
D (hours)	6	8	9
V (visits)	6	10	12

Table 3. Classification of Cost (C), Service Time (D), and Visit (V) Restrictions by Capacity

Factor C (retrieved from *Verbraucherpreisindex für Deutschland*):

The baseline fuel consumption costs for the vehicle categories (€0.25/km for the Low category, €0.38/km for the Medium category, and €0.56/km for the High category) are derived from operational data published by the German Federal Office for Logistics and Mobility in its periodic Verbraucherpreisindex für Deutschland reports.

These values are calculated by crossing the industry-standard fuel consumption rates for European fleets (averaging 14L, 22L, and 33L per 100 km, respectively) with the net commercial diesel prices in Germany, as reported by the Federal Statistical Office (Destatis).

These parameters provide a realistic baseline for the Rich Vehicle Routing Problem (RVRP) objective function, as they reflect actual operating conditions and fuel price standards within the German freight transport sector.

Factor D and V:

Service time restrictions (6, 8, and 9 hours) are aligned with the daily driving limits established under Regulation (EC) No. 561/2006. Likewise, the maximum number of stores per vehicle type (6, 10, and 12, respectively) is not an arbitrary parameter, but an operational constraint derived from time capacity:

Standard Activity Time (T_{total})

For this optimization model, the total time for each store (T_{total}) in minutes is composed of a fixed time, independent of the vehicle size, and a variable time depending on the pallet volume that it is carrying.

Sequence	Activity	Estimated Net time (minutes)	Margin of error (minutes)	(T_{total})	Operative justification
1	Parking and Maneuvering	10	+5 (yard traffic/waiting)	15	Time from when the truck crosses the gate until it is perfectly docked at the loading bay.
2	Paperwork and Opening	5	+5 (Searching for the supervisor)	10	Breaking security seals, the supervisor's signature is on the delivery note.
3	Physical Unloading	1 p/pallet	+0.5 (stuck/misaligned pallet)	1.5	Use of an electric pallet jack.
4	Empty Pallet Collection	0.5 p/pallet	+0.5 (Pallet inspection)	1	Loading empty pallets.

Table 4. Breakdown of standard activity times, including margins of error in minutes.

Service time (T_{si})	Activity description	Example	Time (minutes)
---------------------------	----------------------	---------	----------------

Variable time ($t_{Descharge} + t_{inspection}$) : (1.5 + 1) Volume dependent	This involves the visual and physical quality control of incoming merchandise.	Damage pallet verification.	($T_{Descharge} + T_{inspection}$) • X_i
Fixed time (T_F) Volume independent	Administrative inspection process.	Signing off on the warehouse management system.	25

Table 5. Classification of Total Time Components into Fixed and Variable Activities.

$$\text{Service time } (T_{si}) = T_F + (T_{Descharge} + T_{inspection}) \cdot X_i$$

Where X_i represents the number of pallets of the specific store, which directly modifies the unloading and inspection time at each node, for example (using a high-capacity vehicle):

- If this truck only delivers 2 pallets, the unloading and inspection time will only take 5 minutes $2 \times (1.5 + 1)$.
- If that same truck goes to another node and unloads 20 pallets instead, the worker will take 50 minutes $20 \times (1.5 + 1)$.

Another limitation for the maximum capacity in terms of euro pallets is defined by the following inequality:

$$\sum_{\text{Stores } i \in \text{Route}_k} X_i \leq Q_k$$

This constraint implies that the sum of all pallets from the stores visited by the truck k cannot exceed the maximum capacity Q_k of its vehicle type (*indicated in Table 2*).

IV. Mathematical Model

Decision variables

- Type of vehicle (low-, medium-, or high-capacity) according to the delivery route.
- The sequence in which the Lidl stores should be visited.
- Consequently, the number of vehicles and drivers required for the distribution process.

To support a more sustainable approach, additional decision variables are also considered, while keeping the main objective in focus.

- CO2 emissions generated by the selection of a vehicle category.
- Number of drivers (In accordance with recovery times).

Sets and Indices

Set	Index	Description
N	i, j	Total system nodes. Including all the existing nodes in the area (1 distributor, Dr. Oetker, 24 Lidl stores): $N = \{0, 1, 2, 3 \dots n\}, \text{ where } n = 25$
S	i, j	This subset isolates the stores from the central depot (node 0) within the global node set N . This subset is essential because the depot operates under entirely different logical constraints than the commercial destinations. While stores exhibit specific material demands ($X_i > 0$). This prevents the solver from generating redundant loops and increases efficiency by eliminating unnecessary “if” conditionals. $S = N \setminus \{0\} \text{ or } S = \{1, 2, 3 \dots n\}, \text{ where } n = 24$ This implies that the set excludes the origin node (Node 0).
K	k	A set of available vehicles inside the heterogeneous fleet, where $K = \{1, 2 \dots K\}$
M	m	Type (Category) of vehicle $M = \{ "Low", "Medium", "High" \}$
K_m	$m \in M$	A disjoint subset of individual vehicles belonging strictly to the capacity category mentioned before. $\bigcup_{m \in M} K_m = K \quad K_m \cap K_{m'} = 0, \forall m \neq m'$ $\bigcup_{m \in M}$: Each and every truck in the fleet (K) belongs to one of the three capacity categories. There's no truck in the system considered left out of parameter restrictions $\cap = 0$: An individual truck cannot belong to two categories at once. It is physically impossible for a vehicle to belong to multiple categories simultaneously.

Parameters

d_{ij} : Distance in kilometers (km) between node i and node j . It is assumed that d_{ii} or $d_{jj} = 0$.

t_{ij} : Time of travelling in minutes (min) between node i and node j . Calculated based on the average speed of the road network in the area.

α : (Buffer-Traffic) Road safety and congestion scaling coefficient in the OWL region ($\alpha = X_i$:

Store demand. 1.15)

X_i : Discrete supply demand for each Lidl store $i \in S$ measured in quantity of pallets (units).

For the central depot evidently, is $X_0: 0$

v_{im} : Accessibility matrix (binary):

- 1 = If the vehicle m has physical access to the store $i \in S$
- 0 = If the vehicle is prohibited from accessing to that store i .

Q_m : Maximum pallet capacity ($Q_{Low} = 15$, $Q_{Medium} = 24$, $Q_{High} = 33$) for type of vehicle.

$S_{max m}$: The maximum number of stores that the type of vehicle m can visit in a single route.

u_{ik} : It is a continuous variable representing the number of pallets remaining in truck k . immediately after leaving store i .

- 1 = If the vehicle m has physical access to the store i .
- 0 = If the vehicle is prohibited from accessing to that store i

For example, $v_{im} = 0$ would be when a high-capacity truck cannot have access to the historical center.

$D_{max m}$: Maximum route duration, depending on the driver's shift and type of vehicle m expressed in minutes ($D_{Low} = 360$, $D_{Medium} = 480$, $D_{High} = 540$)

c_m : Variable gas cost in euros (€/km) depending on type of truck and emission penalty measured in euros per kilometer traveled. $c_{Low} = 0.25$, $c_{Medium} = 0.38$, $c_{High} = 0.56$

T_f : Fixed time of service for the administrative procedure. $T_f = 25$ in minutes.

$t_{discharge}$: Variable time in minutes for pallet discharge. (1.5 min/pallet).

$t_{inspection}$: Variable time in minutes for pallet inspection. (1 min/pallet).

Variable decision parameters:

- $y_{ijk} : \{0, 1\}$ Binary variable for the route.
 - 1 = If the vehicle k goes directly from node i to node j .
 - 0 = Otherwise
- u_{ik} : Continuous variable of the subroute. Indicates the order of visit sequence of the store i by the truck k . $u_{ik} \geq 0$ to prevent isolated loops.
- $w_k : \{0, 1\}$ Binary fleet usage variable.
 - 1 = If the vehicle k is activated to work.
 - 0 = If it is stored in the main deposit.

Objective Function

The goal of this mathematical function is to minimize the variable financial cost of distribution, which directly weights the environmental impact (CO2 emissions) resulting from the fuel consumption of the heterogeneous fleet over the distances traveled in direct proportion.

$$Min Z = \sum_{i \in N} \sum_{j \in N, j \neq i} \sum_{m \in M} \sum_{k \in K_m} c_m \cdot d_{ij} \cdot y_{ijk}$$

Breakdown:

1. $Min Z$: Optimization criterion is the minimization, which represents the economic and environmental cost of daily operation.
2. $\sum_{i \in N} \sum_{j \in N, j \neq i}$: Double summation over the network's adjacency matrix that checks every possible trip from origin node i to a destination node j . The condition $j \neq i$ ensures that a node cannot be treated as its own destination, preventing optimization from creating self-loops or assigning travel costs from a node to itself. Within the summation, the multiplication by the adjacency matrix acts as a filter: only origin–destination pairs connected by an existing network link contribute to the objective function, while non-connected pairs contribute zero. The double summation then aggregates the contribution of all valid origin–destination pairs across the network.
3. $\sum_{m \in M} \sum_{k \in K_m}$: Double summation for the heterogeneous fleet. It checks that each category of vehicle m (Low, Medium, High), and sequentially, each individual physical vehicle k belongs to the subset K_m .
4. c_m : Weighted Cost Coefficient, which not only represents the financial cost of fuel per kilometer but also proportionally penalizes the CO2 emissions factor.
5. d_{ij} : Physical distances in kilometers between nodes in the OWL region.
6. y_{ijk} : A routing variable that acts as the “switch” in the equation. If the truck k of type m travels within i and j , $y_{ijk} = 1$, the cost of that segment is added to the total ($c_m \cdot d_{ij} \cdot 1$). If the truck does not travel that distance, $y_{ijk} = 0$, nullifying the product and preventing any cost from being added to the system.

System Constraints

1. $\sum_{j \in S} y_{0jk} = w_k \quad \forall m \in M, \forall k \in K_m$
2. $\sum_{i \in S} y_{0ik} = w_k \quad \forall m \in M, \forall k \in K_m$
3. $\sum_{i \in N, i \neq 1} \sum_{k \in K} y_{ijk} = 1 \quad \forall j \in S$

4. $\sum_{j \in N, j \neq 1} y_{ijk} - \sum_{h \in N, h \neq j} y_{ijk} = 0 \quad \forall j \in S, \forall k \in K$
5. $\sum_{j \in N, j \neq 1} y_{ijk} \leq v_{jm} \quad \forall j \in S, \forall m \in M, \forall k \in K_m$
6. $\sum_{j \in S} \sum_{i \in N, i \neq j} X_j \cdot y_{ijk} \leq Q_m \cdot w_k \quad \forall m \in M, \forall k \in K_m$
7. $\sum_{j \in S} \sum_{i \in N, i \neq j} y_{ijk} \leq S_m^{max} \cdot w_k \quad \forall m \in M, \forall k \in K_m$
8. $\sum_{i \in N} \sum_{j \in N, j \neq i} (\alpha \cdot t_{ij}) \cdot y_{ijk} + \sum_{j \in S} (T_f + (t_{discharge} + t_{inspection}) \cdot X_j) \cdot (\sum_{i \in N, i \neq j} y_{ijk}) \leq D_m^{max} \cdot w_k$
 $\forall m \in M, \forall k \in K_m$
9. $u_{ik} - u_{jk} + Q_m \cdot y_{ijk} \leq Q_m - X_j \quad \forall i \in N, \forall j \in N, \forall j \in S, i \neq j, \forall m \in M, \forall k$
10. $0 \leq u_{ik} \leq Q_m \cdot w_k \quad \forall i \in N, \forall m \in M, \forall k \in K_m$

Variables:

$y_{ijk} \in \{0, 1\} \quad \forall i, j \in N, \forall k \in K$	It constrains the algorithm's search space to prevent absurd solutions from the solver, such as using "0.5 vehicles or -1 trips" between supermarkets.
$w_k \in \{0, 1\} \quad \forall k \in K$	Therefore, routing and fleet usage decisions are binary, whereas the sequential sub-route counters are positive continuous variables.
$u_{ik} \geq 0 \quad \forall i \in N_0, \forall k \in K$	

Indices

$i \in N$: Index representing the origin node of a route (which may be the distribution center (DC) or a Lidl store).

$j \in N$: Index representing the intermediate destination node or arrival stop visited by a vehicle.

$h \in N$: Secondary index representing the subsequent destination node (departure node) to which a vehicle travels immediately after completing a delivery.

Explanation of System Constraints

R1: Mandatory exit from the Central Depository

Origin node is fixed in $i = 0$ (Dr. Oetker Central Depository). The summation evaluates all possible destinations of the set of store, and the same equation is repeated for each vehicle k inside the heterogeneous fleet K_m .

- If vehicle k is activated ($w_k=1$) , consequently $\sum_{j \in S} y_{0jk} = 1$.
- If the vehicle is not used ($w_k = 0$), the restriction means that the truck cannot enter or exit from the deposit.

R2. Mandatory Return to the Central Depository

The destination node is fixed in $j = 0$ (Dr. Oetker Central Depository). The summation evaluates all possible origin nodes of the set of stores from which the vehicle can return.

- The truck is required to close its cycle by generating one arc leading back to the central depot. This ensures that no route remains open once deliveries are complete.

R3: Total Coverage and Single Customer Visit

By setting this constraint strictly equal to 1, the solution space is restricted so that each store is visited exactly once by a single truck. If an inequality (≥ 1) were used instead, the model would allow two different trucks to visit the same store to split the order, which would violate Lidl's logistics policy of avoiding duplicate loading dock costs and administrative paperwork.

R4: Flow Conservation at Commercial Nodes

It is evaluated for every combination of store $j \in S$ and vehicle $k \in K$. The index i represents the nodes entering j , while the index h represents the nodes departing from j .

- **Active Transit (1-1= 0)** : If vehicle k is assigned to serve store j from an origin node i , the incoming binary variable is activated ($y_{ijk} = 1$). To satisfy the equality to zero, the solver is then forced to activate exactly one outgoing binary variable ($y_{jhk} = 1$). This ensures route continuity by requiring the vehicle to proceed to a subsequent destination node h .
- **Inactive (0 - 0 = 0)**: If the optimal route of vehicle k does not include store j , all binary variables associated with that node take the value zero. The flow balance is therefore satisfied by default, allowing the vehicle to travel through other regions of the network without affecting the connectivity matrix of the unvisited nodes.

R5: Accessibility physical restriction

- If store j (for example, in an OWL-restricted historic city center) prohibits high-capacity trucks, the parameter is set to $v_{j"High"} = 0$. The inequality then becomes ≤ 0 , which immediately eliminates any inbound arc to that store served by a heavy vehicle.
- If the store has permitted access $v_{jm} = 0$, the inequality then becomes ≤ 1 , making the solver freely decide if that route can be used or not.

R6: Fleet Maximum Load Capacity Constraint

This constraint prevents the physical overloading of a vehicle.

- If truck k visits store j ($y_{ijk} = 1$), the store's pallet demand is accounted for $X_j \cdot 1$. The summation, therefore, accumulates the total pallet demand of all stores assigned to the route of that specific truck.
- If the truck is inactive ($w_k = 0$), the right-hand side becomes 0, ensuring that no load is assigned to it. If the truck is active ($w_k = 1$), the upper bound is strictly set to the nominal capacity of its corresponding vehicle category $Q_m = (15, 24, 33 \text{ pallets})$.

R7: Commercial Stop Limit per Route Constraint

This constraint counts a value of "1" each time an arc leading to a commercial store is activated. The sum of all inbound commercial arcs, therefore, represents the exact number of delivery stops assigned to a route.

- By bounding this quantity with $S_m^{max} \cdot w_k$, the solver is prevented from generating excessively long routes that could lead to driver fatigue or compromise product freshness. The maximum allowable number of stops is adjusted according to the corresponding truck category m .

R8: Traffic-Adjusted Workday Time Window

- If truck k visits store j , the corresponding assignment variable equals 1, activating the service time at that location. The service time consists of a fixed administrative component plus a variable unloading and inspection time that depends on the number of pallets delivered to the store. This service time is added to the total driving time of the route.
- To account for traffic congestion in the OWL region, all travel times are multiplied by a safety factor of $\alpha=1.15$, increasing the nominal travel times by 15%. The resulting total route duration (including both driving and service times) must not exceed the maximum allowable working time for the driver. This upper bound is enforced only when the truck is active ($w_k=1$); otherwise, the right-hand side becomes zero, ensuring that an inactive vehicle cannot be assigned a route.

R9: Load Propagation Constraint

The continuous variable tracks the amount of cargo remaining on a truck as it moves along its route.

- If a connection between two locations is not part of the selected route, this constraint has no practical effect and does not influence the solution.

However, when the truck actually travels from one location to the next, the constraint ensures that the remaining load is updated correctly. After serving a store, the truck must carry fewer

pallets than before, with the reduction matching exactly the number of pallets delivered at that stop.

As a result, the truck's load decreases progressively along the route, guaranteeing that deliveries are made consistently and that the remaining cargo is always physically feasible. This prevents the model from creating unrealistic routes in which the truck's load does not change after completing deliveries.

R10: Residual Load Boundary Conditions

This constraint defines the valid range of the residual load carried by a truck at any point along its route. The remaining load can never be negative, since a truck cannot deliver more pallets than it carries. Likewise, it can never exceed the truck's maximum loading capacity.

- If a truck is not used, the constraint forces its residual load to remain zero throughout the model. Consequently, only active trucks are allowed to carry cargo, while inactive trucks are assigned no load.

Solver Approach

Solver and version: *

Model Type: Mixed-Integer Linear Programming (MILP)

The model is classified as mixed because it combines binary decision variables, which determine whether a truck travels between two locations, with continuous variables, which represent the amount of cargo remaining on each truck throughout its route (Fachrizal, 2020). The problem is computationally complex because it simultaneously involves deciding how to assign customer demands to vehicles while determining the optimal sequence of deliveries. These decisions combine characteristics of the *Bin Packing Problem* and the *Traveling Salesman Problem*.

IMPORTS

```
import json
import time
import os
import random
import copy
import math
```

```
import pulp
from collections import defaultdict
```

PROTOTYPE: (WE CAN USE THIS AS AN SKETCH)
And from this sketch, we MUST put the constraints**
Class:

Location planning in Python /PuLP
Data from previous example

```
customers = ("v", "w", "x", "y", "z")
locations = ("A", "B", "C", "D")

transport_costs = \
{("A", "v"): 39, ("A", "w"): 31, ("A", "x"): 49, ("A", "y"): 15, ("A", "z"): 40,
 ("B", "v"): 56, ("B", "w"): 45, ("B", "x"): 25, ("B", "y"): 24, ("B", "z"): 47,
 ("C", "v"): 26, ("C", "w"): 48, ("C", "x"): 40, ("C", "y"): 16, ("C", "z"): 50,
 ("D", "v"): 40, ("D", "w"): 50, ("D", "x"): 41, ("D", "y"): 28, ("D", "z"): 52}

fixed_costs = {"A": 85,
               "B": 120,
               "C": 25,
               "D": 30}
```

```
import json
import time
import os
import random
import copy
import math
import pulp
from collections import defaultdict
```

```
# -----
# 1. Solutions (Data Structure)
# -----
class Solution:
    def __init__(self, routes=None):
        self.routes = routes if routes else {}
        self.cost = float("inf")
        self.feasible = False

    def copy(self):
        return Solution(copy.deepcopy(self.routes))

# -----
```

```

# 2. COST FUNCTION (Paper Evaluation with Window Times)
# -----
def evaluate(solution, data, capacity, fixed_cost, var_cost):
    dist = data["distance_matrix"]
    time_matrix = data["travel_time_matrix"]
    demand = {int(k): v for k, v in data["demands"].items()}
    customers = set(int(c) for c in data["customer_ids"])

    tw_start = {int(k): v[0] for k, v in data["time_windows"].items()}
    tw_end = {int(k): v[1] for k, v in data["time_windows"].items()}
    service_time = {int(k): v for k, v in data["service_times"].items()}

    depot_end = max(customers) + 1

    # Heredar ventanas del depósito inicial para el nodo virtual de llegada
    tw_start[depot_end] = data["time_windows"]["0"][0]
    tw_end[depot_end] = data["time_windows"]["0"][1]
    service_time[depot_end] = 0

    total = 0
    visited = set()
    feasible = True

    for v, route in solution.routes.items():
        if len(route) <= 2:
            continue

        total += fixed_cost[v]
        load = 0
        current_time = tw_start[0]

        for i in range(len(route) - 1):
            a, b = route[i], route[i+1]

            a0 = 0 if a == depot_end else a
            b0 = 0 if b == depot_end else b

            total += dist[a0][b0] * var_cost[v]

            if b in customers:
                load += demand[b]
                visited.add(b)

    # Simulación de propagación temporal (HFVRPTW)

```

```

    arrival_time = current_time + service_time[a0] + time_matrix[a0][b0]
    if arrival_time < tw_start[b0]:
        current_time = tw_start[b0]
    elif arrival_time > tw_end[b0]:
        feasible = False
        total += 1e5 * (arrival_time - tw_end[b0]) # Penalización temporal
        current_time = arrival_time
    else:
        current_time = arrival_time

    if load > capacity[v]:
        feasible = False
        total += 1e6 * (load - capacity[v]) # Penalización de sobrecarga

    missing = customers - visited
    if missing:
        feasible = False
        total += 2e6 * len(missing) # Penalización por tiendas omitidas

    solution.cost = total
    solution.feasible = feasible
    return total, feasible

# -----
# 3. INITIAL SOLUTION
# -----
def initial_solution(data, vehicle_ids, capacity):
    customers = sorted(list(map(int, data["customer_ids"])))
    dist = data["distance_matrix"]
    demand = {int(k): v for k, v in data["demands"].items()}

    depot = 0
    depot_end = max(customers) + 1
    unvisited = set(customers)

    sol = Solution({v: [depot, depot_end] for v in vehicle_ids})

    for v in vehicle_ids:
        if not unvisited:
            break

    load = 0
    route = [depot]

```

```

while True:
    best = None
    best_d = float("inf")

    for c in list(unvisited):
        if load + demand[c] <= capacity[v]:
            d = dist[route[-1]][c]
            if d < best_d:
                best = c
                best_d = d

    if best is None:
        break

    route.append(best)
    load += demand[best]
    unvisited.remove(best)

    route.append(depot_end)
    sol.routes[v] = route

return sol

# -----
# 4. ALNS OPERATORS (Sparsification & Costo Delta de Inserción)
# -----
def destroy_random(sol, pct=0.25):
    s = sol.copy()
    removed = []

    for v, r in s.routes.items():
        if len(r) <= 3:
            continue
        clients = r[1:-1]
        k = max(1, int(len(clients) * pct))
        rem = random.sample(clients, min(len(clients), k))

        for c in rem:
            r.remove(c)
            removed.append(c)

    return s, removed

```



```

def destroy_worst(sol, data, pct=0.25):
    s = sol.copy()
    dist = data["distance_matrix"]
    customers = list(map(int, data["customer_ids"]))
    depot_end = max(customers) + 1

    scores = []
    for v, r in s.routes.items():
        if len(r) <= 3:
            continue
        for i in range(1, len(r) - 1):
            a, b, c = r[i-1], r[i], r[i+1]

            a0 = 0 if a == depot_end else a
            b0 = 0 if b == depot_end else b
            c0 = 0 if c == depot_end else c

            # Costo de ahorro inverso (Remoción limpia)
            score = dist[a0][b0] + dist[b0][c0] - dist[a0][c0]
            scores.append((score, b, v))

    scores.sort(key=lambda x: x[0], reverse=True)
    k = max(1, int(len(scores) * pct))

    removed = []
    for _, c, v in scores[:k]:
        if c in s.routes[v]:
            s.routes[v].remove(c)
            removed.append(c)

    return s, removed

def repair(sol, removed, data, capacity):
    s = sol.copy()
    dist = data["distance_matrix"]
    demand = {int(k): v for k, v in data["demands"].items()}
    depot_end = max(map(int, data["customer_ids"])) + 1

    random.shuffle(removed)

    for c in removed:
        best_cost = float("inf")
        best = None

```

```

for v, r in s.routes.items():
    load = sum(demand[n] for n in r if n in demand)
    if load + demand[c] > capacity[v]:
        continue

    for i in range(1, len(r)):
        a, b = r[i-1], r[i]
        a0 = 0 if a == depot_end else a
        b0 = 0 if b == depot_end else b

        cost = dist[a0][c] + dist[c][b0] - dist[a0][b0]
        if cost < best_cost:
            best_cost = cost
            best = (v, i)

    if best:
        v, i = best
        s.routes[v].insert(i, c)
    else:
        # Fallback forzado de contingencia para preservar viabilidad estructural
        for v, r in s.routes.items():
            r.insert(1, c)
            break

return s

# -----
# 5. METÁFORA DE ACEPTACIÓN & CLUSTERIZACIÓN (Spatio-Temporal Core)
# -----

def accept(new, current, T):
    if new.cost < current.cost:
        return True
    if T <= 0:
        return False
    return random.random() < math.exp((current.cost - new.cost) / T)

def extract_cluster(solution, data, k=8):
    """Extrae un subgrafo de tiendas próximas entre sí espacialmente para optimizar."""
    pool = []
    for r in solution.routes.values():
        pool += [n for n in r if n not in (0, max(r))]

```

```

if not pool:
    return []

base_node = random.choice(pool)
dist = data["distance_matrix"]

# Ordenar tiendas Lidl por cercanía al nodo base elegido
pool.sort(key=lambda x: dist[0 if base_node==max(pool)+1 else base_node][0 if
x==max(pool)+1 else x])
return pool[:min(k, len(pool))]

# -----
# 6. MILP LOCAL REFORMULADO (Single-Commodity Flow Exacto Ajustado)
# -----
def milp_local(data, nodes, vehicles, capacity, fixed, var, time_limit=12):
    if not nodes or len(nodes) < 2:
        return None

    depot = 0
    # Usar un identificador virtual local que no colisione con los índices del JSON
    depot_end = 999

    local_nodes = [depot] + sorted(list(set(nodes))) + [depot_end]
    demand = {int(k): v for k, v in data["demands"].items()}
    demand[depot] = 0
    demand[depot_end] = 0

    model = pulp.LpProblem("Local_Subproblem_SCF", pulp.LpMinimize)

    # Variables binarias y continuas indexadas sobre la red local
    x = pulp.LpVariable.dicts("xl", ((i,j,k) for i in local_nodes for j in local_nodes for k in
vehicles if i!=j), cat="Binary")
    f = pulp.LpVariable.dicts("fl", ((i,j,k) for i in local_nodes for j in local_nodes for k in
vehicles if i!=j), lowBound=0, cat="Continuous")

    # Mapeo de costos reales reconduciendo índices virtuales al Depot Inicial (0) del JSON
    model += pulp.lpSum(
        var[k] * data["distance_matrix"][0 if i==depot_end else i][0 if j==depot_end else j] *
x[i,j,k]
        for i,j,k in x
    )

    # Cobertura obligatoria de tiendas Lidl del clúster
    for j in local_nodes:

```

```

    if j not in (depot, depot_end):
        model += pulp.lpSum(x[i,j,k] for i in local_nodes for k in vehicles if i!=j) == 1

# Conservación de Flujo de Ruta Tradicional
for k in vehicles:
    model += pulp.lpSum(x[depot, j, k] for j in local_nodes if j != depot) <= 1
    model += pulp.lpSum(x[i, depot_end, k] for i in local_nodes if i != depot_end) <= 1
    for p in local_nodes:
        if p not in (depot, depot_end):
            model += (pulp.lpSum(x[i,p,k] for i in local_nodes if i!=p) ==
                      pulp.lpSum(x[p,j,k] for j in local_nodes if j!=p))

# REFORMULACIÓN MATEMÁTICA SCF CORRECTA: El flujo neto consume la
demanda real de palets
for k in vehicles:
    for i in local_nodes:
        if i not in (depot, depot_end):
            flujo_entrante = pulp.lpSum(f[j,i,k] for j in local_nodes if j!=i)
            flujo_saliente = pulp.lpSum(f[i,j,k] for j in local_nodes if j!=i)
            model += (flujo_entrante - flujo_saliente == demand[i] * pulp.lpSum(x[i,j,k] for j
in local_nodes if j!=i))

# Acotación fuerte de capacidad por arco
for k in vehicles:
    for i,j in x:
        model += f[i,j,k] <= capacity[k] * x[i,j,k]

solver = pulp.PULP_CBC_CMD(timeLimit=time_limit, msg=0, presolve=True)
model.solve(solver)

if pulp.LpStatus[model.status] != "Optimal":
    return None

# Mapear de regreso los nodos virtuales al índice real global del depósito de salida
global_depot_end = max(map(int, data["customer_ids"])) + 1
routes = {}

for k in vehicles:
    if pulp.lpSum(x[depot, j, k] for j in local_nodes if j != depot).value() > 0.5:
        route = [depot]
        curr = depot
        visited_sub = set()
        while True:
            if curr in visited_sub:

```

```

        break
    visited_sub.add(curr)
    nxt = [j for i,j,kk in x if i==curr and kk==k and pulp.value(x[i,j,kk]) > 0.5]
    if not nxt:
        break
    curr = nxt[0]
    route.append(global_depot_end if curr == depot_end else curr)
    if curr == depot_end:
        break
    if len(route) > 2:
        routes[k] = route

return routes

# -----
# 7. MAIN HYBRID ALNS + MILP TRACKER
# -----
def run_hybrid(data, vehicle_ids, capacity, fixed, var, iters=100):
    print(f'\n{'='*65}')
    print(f' MATHEURÍSTICO (ALNS + MILP SCF)')
    print(f'\n{'='*65}')

    sol = initial_solution(data, vehicle_ids, capacity)
    evaluate(sol, data, capacity, fixed, var)

    best = sol.copy()
    T = 1200
    global_depot_end = max(map(int, data["customer_ids"])) + 1

    for it in range(1, iters + 1):
        if random.random() < 0.5:
            s2, rem = destroy_random(sol, pct=0.20)
        else:
            s2, rem = destroy_worst(sol, data, pct=0.20)

        s2 = repair(s2, rem, data, capacity)
        evaluate(s2, data, capacity, fixed, var)

        if accept(s2, sol, T):
            sol = s2

    if s2.feasible and s2.cost < best.cost:
        best = s2.copy()
        print(f' Iter {it:03d} | ¡Mejora ALNS Global! -> Costo: {best.cost:.2f} €')

```

```

# INTENSIFICACIÓN MILP LOCAL EXACTA POR CLÚSTER
if it % 15 == 0:
    cluster = extract_cluster(sol, data, k=8)
    # Utilizar los vehículos asignados al subset para correr el solver de plano de corte
    improved = milp_local(data, cluster, vehicle_ids[:3], capacity, fixed, var,
time_limit=8)

    if improved:
        # Limpiar la secuencia vieja eliminando los duplicados
        for v in vehicle_ids:
            if v in sol.routes:
                sol.routes[v] = [n for n in sol.routes[v] if n not in cluster]

        # Sincronizar e inyectar sub-rutas óptimas
        sol.routes.update(improved)

        # Reparar extremos rotos por vaciado absoluto
        for v, r in sol.routes.items():
            if len(r) == 0 or r == [0]:
                sol.routes[v] = [0, global_depot_end]

        evaluate(sol, data, capacity, fixed, var)
        if sol.feasible and sol.cost < best.cost:
            best = sol.copy()
            print(f" Iter {it:03d} || -> Costo: {best.cost:.2f} €")

T *= 0.993

print(f"\n{'='*65}")
print(f" Costo Óptimo de Investigación : {best.cost:.2f} €")
print(f" Solución Validada Viable      : {best.feasible}")
print(f"{'='*65}")
return best

```

Heuristic Approach

For the construction of the parameterized heuristic, three classic route initialization methodologies were evaluated: sequencing by radial proximity (closeness to the central depot), decreasing ordering by demand criticality (servicing type-L demands first and filling capacity with type-S), and the approach based on the maximization of the savings function (Clarke & Wright Savings Algorithm).

While demand-based and proximity-based approaches optimize fleet utilization and depot outbound arcs respectively, the final model employs a savings matrix integrated with a geographical accessibility filter. This methodology is justified by its dual optimization capability: it simultaneously leverages the geometric efficiency of consolidating adjacent nodes and strictly enforces store-specific time windows. Consequently, this prevents the algorithmic isolation of peripheral, high-demand nodes during the final stages of route generation

Three Classic Methods for Routing Initialization and Service Sequencing

A. By proximity to the central depot (Proximity-Based Approach)

- **How it works:** Priority is given to servicing the nodes closest to the depot (Node 0, Dr. Oetker) first, or grouping them by geometric/radial sectors from the center outward.
- **Advantage:** It intuitively minimizes the initial travel times outward from the depot.
- **Disadvantage:** It can leave more distant stores isolated during the final stages of the algorithm. If the larger trucks run out of capacity, you will be forced to dispatch a dedicated vehicle to a single remote location, causing overall logistics costs to skyrocket (logistics bullwhip effect).

B. By demand volume (Criticality-Base Approach)

- **How it works:** Stores are sorted in descending order of demand (d_j). Large stores (Lidl type-L, requesting between 14 and 25 pallets) are assigned first to the trucks with the highest capacity (Q_m). Once these "large blocks" are allocated, small demands (Lidl type-S, between 2 and 5 pallets) are used to fill the remaining empty spaces in the vehicles.
- **Advantage:** It ensures that the most challenging locations with severe accessibility restrictions are resolved from the beginning.
- **Disadvantage:** It temporarily ignores geographical distances, which could lead to grouping two high-demand stores located at opposite extremes of the Ostwestfalen-Lippe (OWL) region.

C. By the "Savings" concept (Clarke & Wright Algorithm)

Given that a high-quality, custom heuristic is required for this project, industry standards do not initiate routing based solely on proximity or demand. Instead, they compute a Savings Matrix (SS_{ij}).

- **How it works:** The algorithm initially assumes that each store is visited by an individual truck that travels back and forth from the depot ($0 \rightarrow i \rightarrow 0$). Then, it calculates how much travel is "saved" if two stores are combined into the same route ($0 \rightarrow i \rightarrow j \rightarrow 0$).

- Mathematical formula: $S_{ij} = c_{0i} + c_{0j} - c_{ij}$ (where c represents travel distance or time).
- Ordering criterion: Routing sequencing begins by linking the two nodes that generate the highest possible time/distance savings, provided that time windows and the assigned truck's capacity constraints are not violated.

Recomendaciones para asegurar el 100% en la entrega:

Asegúrate de que el código de la heurística refleje fielmente la explicación teórica:

1. **El Filtro de Accesibilidad (v_{jm}):** Cuando el algoritmo intente fusionar la ruta de la tienda i y la tienda j para asignarlas a un camión tipo m , el código debe verificar primero si el camión m tiene permitido el acceso a ambas tiendas según tu matriz de restricciones logísticas.
2. **Capacidad de Carga Flota Heterogénea:** Como tu flota es heterogénea, el cálculo del ahorro debe ser dinámico. Si combinas la demanda de i y j , debes validar si la suma de paletas cabe en el camión disponible más pequeño que tenga acceso a esos nodos.
3. **El Formato de Comparación:** En tu sección de resultados, asegúrate de hacer una tabla comparativa directa (instancia pequeña, mediana y grande) que mida:
 - **Función Objetivo:** Cuánto tiempo/costo total obtuvo Gurobi vs. la Heurística.
 - **Tiempo de Cómputo (Runtime):** Mostrar que la heurística resuelve la instancia grande en milisegundos, mientras que el solver sufre o agota el tiempo límite (Stress Test). Esto cerrará con broche de oro la justificación de por qué desarrollaste la heurística.

Trade-offs

The optimal solution is not straightforward because several real-world factors create conflicting objectives and limitations within the routing and fleet assignment problem:

- High-capacity vehicles may reduce the number of drivers required but could increase unloading times or reduce route flexibility.
- Smaller vehicles may improve maneuverability and scheduling flexibility, although they require more drivers and trips.

- The shortest geographical distance between two points is not always the fastest route, since real road networks are not linear and include turns, intersections, and indirect paths.
- Routes that appear shorter may pass through highly congested roads, leading to longer travel times compared to longer but faster alternative routes.
- Certain streets may be too narrow or restricted for specific vehicle types, making some theoretically shortest routes infeasible.
- Road infrastructure constraints such as toll roads, traffic signals, pedestrian zones, or delivery restrictions can affect route feasibility and cost.
- Delivery time windows can force vehicles to take longer detours in order to arrive at stores at a feasible time.
- Loading and unloading constraints, including ergonomic limitations, may increase service time at certain stops and influence route sequencing decisions.
- Traffic variability during the day means that the same route can have different travel times depending on departure time.

Because of these interacting factors, the “optimal” route is not simply the shortest or low-cost one, yet the one that achieves the best balance between time, cost, feasibility, and operational constraints.

Appendices

App. 1

stochastic: “A stochastic process or system is connected with random probability” (Cambridge Dictionary, n.d).

App. 2. OWL area:



Image retrieved from:

https://www.amazon.de/s?k=reisef%C3%BChrer+ostwestfalen+lippe&adgrpid=1206164911060879&hvadid=75385424524280&hvbmt=be&hvdev=c&hvlocint=116641&hvlocphy=129550&hvnetw=o&hvqmt=e&hvtargid=kwd-75385511156668%3Aloc-72&hydadcr=29569_241

[3232&mcid=2fea2a13ec653ea2a1b8eede075b618c&msclkid=a9fed498ddeb117b1bdbbeab726c5623&tag=hyddemsn-21&ref=pd_sl_956js848nk_e\)](https://doi.org/10.1007/978-0-387-77778-8_1)

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